8 Required experiments

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Run	Embedding condition	Embedding layer	Purpose	Best configuration	Results
E0	Random 100-dimensional vectors	Frozen or fixed control	Measure what the LSTM learns without external semantic initialization.	Frozen embedding; LSTM 1x128, Adam 1e-3, batch 64, patience 2 best epoch 5 / 15 / 4 (seeds 0/1/2); mean val loss 0.601	**Completed - seeds 0, 1, 2** (mean ± std) Accuracy: **67.92% ± 4.77** Precision / Recall: 0.7130 ± 0.0571 / 0.6000 ± 0.0471 F1 (positive): 0.6516 ± 0.0516 Macro-F1: **0.6772 ± 0.0480** ROC-AUC: **0.7427 ± 0.0632**
E1	Word2Vec CBOW trained on IMDB training text	Frozen	Test word-level local context representations.	Frozen embedding; LSTM 1x128, Adam 1e-3, batch 64, patience 2 best epoch 8 / 6 / 7 (seeds 0/1/2); mean val loss 0.313	**Completed - seeds 0, 1, 2** (mean ± std) Accuracy: **87.29% ± 0.43** Precision / Recall: 0.8762 ± 0.0100 / 0.8687 ± 0.0078 F1 (positive): 0.8724 ± 0.0038 Macro-F1: **0.8729 ± 0.0043** ROC-AUC: **0.9448 ± 0.0017**
E2	GloVe 100-dimensional public vectors	Frozen	Test global co-occurrence representations from an external corpus.	Frozen embedding; LSTM 1x128, Adam 1e-3, batch 64, patience 2 best epoch 12 / 8 / 8 (seeds 0/1/2); mean val loss 0.319	**Completed - seeds 0, 1, 2** (mean ± std) Accuracy: **86.68% ± 0.33** Precision / Recall: 0.8690 ± 0.0101 / 0.8642 ± 0.0210 F1 (positive): 0.8664 ± 0.0056 Macro-F1: **0.8668 ± 0.0033** ROC-AUC: **0.9411 ± 0.0026**
E3	fastText CBOW trained on IMDB training text	Frozen	Test the value of character n-grams and subword information.	Frozen embedding; LSTM 1x128, Adam 1e-3, batch 64, patience 2 best epoch 6 / 5 / 6 (seeds 0/1/2); mean val loss 0.334	**Completed - seeds 0, 1, 2** (mean ± std) Accuracy: **86.40% ± 0.06** Precision / Recall: 0.8625 ± 0.0159 / 0.8666 ± 0.0228 F1 (positive): 0.8643 ± 0.0036 Macro-F1: **0.8639 ± 0.0006** ROC-AUC: **0.9352 ± 0.0025**
E4	Word2Vec, GloVe, and fastText repeated	Trainable	Measure how much task-specific fine-tuning changes the comparison.	Trainable embedding; LSTM 1x128, Adam 1e-3, batch 64, patience 2 **Word2Vec:** best epoch 4 / 3 / 4 (seeds 0/1/2); mean val loss 0.330 **GloVe:** best epoch 3 / 3 / 3 (seeds 0/1/2); mean val loss 0.342 **fastText:** best epoch 6 / 5 / 4 (seeds 0/1/2); mean val loss 0.323 **Best (lowest val loss): fastText**	**Completed - seeds 0, 1, 2 (mean ± std)** **Word2Vec** Accuracy: **85.87% ± 0.58** Precision: 0.873 ± 0.013 Recall: 0.840 ± 0.025 F1 (positive): 0.856 ± 0.008 Macro-F1: **0.859 ± 0.006** ROC-AUC: **0.932 ± 0.004** **GloVe** Accuracy: **84.48% ± 0.32** Precision: 0.890 ± 0.024 Recall: 0.788 ± 0.027 F1 (positive): 0.835 ± 0.006 Macro-F1: **0.844 ± 0.003** ROC-AUC: **0.928 ± 0.005** **fastText** Accuracy: **86.55% ± 0.24** Precision: 0.867 ± 0.006 Recall: 0.864 ± 0.013 F1 (positive): 0.865 ± 0.004 Macro-F1: **0.866 ± 0.002** ROC-AUC: **0.937 ± 0.002**

Evaluation: Official IMDB test set, 25,000 reviews; one-layer unidirectional LSTM (128 hidden units).

Selection: Best epoch chosen by validation loss (best epochs observed: 3-15); early stopping patience 2. Seeds: 0, 1, 2.

Source: results/metrics.csv (per run) and results/summary.csv. Precision, recall and positive-class F1 use label 1; threshold = 0.5.